

HOUSE PRICE PREDICTION – PROJECT SUMMARY

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, parking availability, furnishing status, and location-related amenities. The project aimed to help understand which factors have the greatest influence on property value and evaluate the effectiveness of different regression models.

Dataset

The Housing Prices Dataset from Kaggle was used for this analysis. The dataset contains 545 records and 13 features describing different aspects of residential properties. The target variable was house price.

Methodology

The dataset was explored using Pandas to understand its structure and identify missing values or duplicates. Categorical variables were converted into numerical format using one-hot encoding. The cleaned dataset was then divided into training and testing sets using an 80:20 split. Two machine learning models were trained and evaluated: Linear Regression and Random Forest Regressor.

Model Performance

Linear Regression achieved the best performance with a Mean Absolute Error (MAE) of approximately 970,043, a Root Mean Squared Error (RMSE) of approximately 1,324,507, and an R^2 Score of 0.653. Random Forest achieved an R^2 Score of 0.613. Based on these results, Linear Regression was selected as the final model because it provided better predictive accuracy for this dataset.

Key Findings

The analysis indicated that factors such as property area, number of bathrooms, parking availability, air conditioning, and preferred area significantly influenced house prices. Larger properties with additional amenities generally commanded higher market values. One interesting observation was that amenities and location characteristics had a strong impact on pricing alongside house size.

Business Recommendation

Real estate businesses should emphasize property size, location advantages, and key amenities when marketing properties. These factors were found to have the greatest influence on house prices and can help improve pricing strategies and customer decision-making.